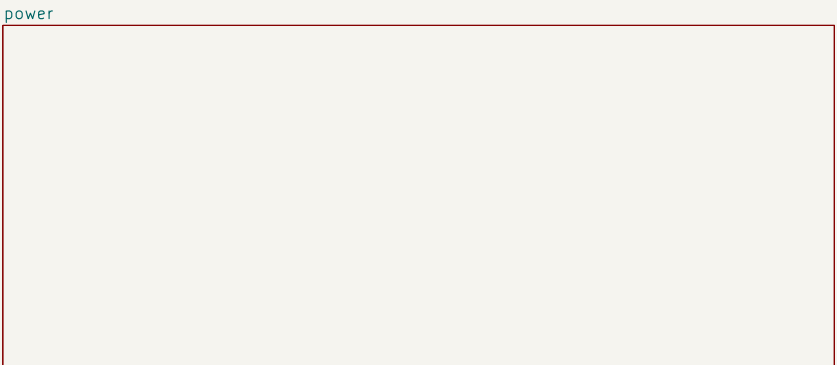


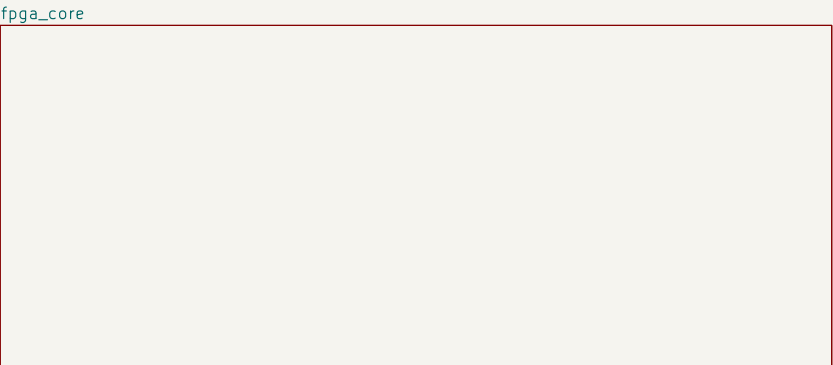
SingleBoard 68K

A J68-derived 68000 soft CPU in an XC6SLX16-FTG256 at 50 MHz, with 32 MiB of MT48LC16M16A2-75 SDRAM and a 3.3 V logic-level UART.

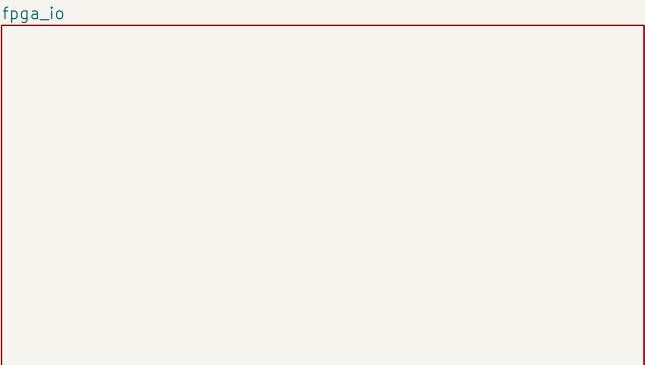
CONCEPT SCHEMATIC, generated by scripts/gen_schematic.py.
Not simulated, not fabricated, not electrically validated.



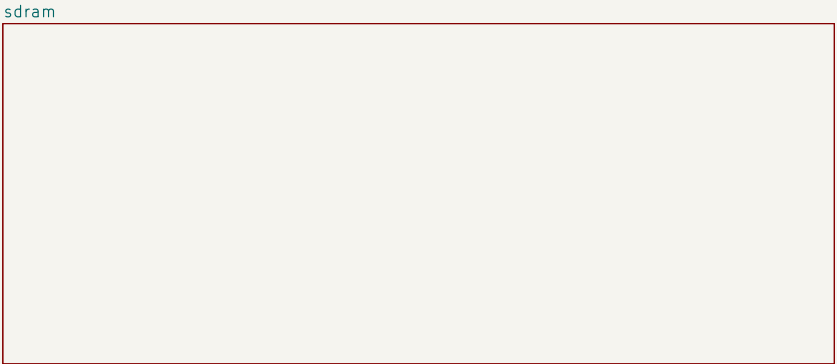
File: power.kicad_sch



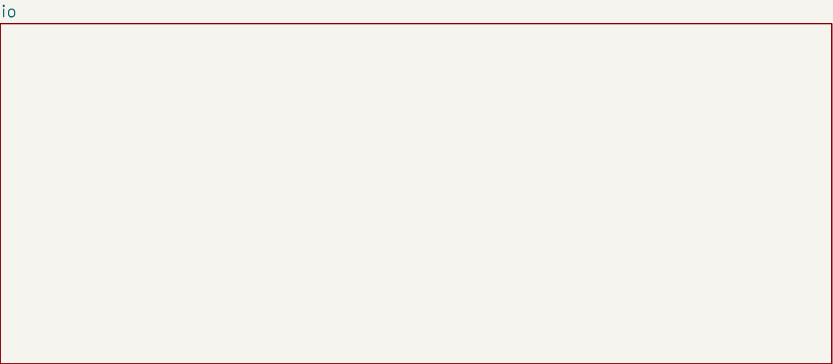
File: fpga_core.kicad_sch



File: fpga_io.kicad_sch



File: sdram.kicad_sch



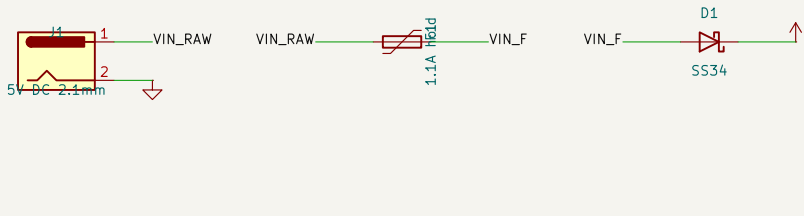
File: io.kicad_sch

Cross-sheet connectivity uses global labels, so the sheets carry no hierarchy pins. Rails are +5V, +3V3 (VCC0, VCCAUX, SDRAM, flash) and +1V2 (VCCINT).

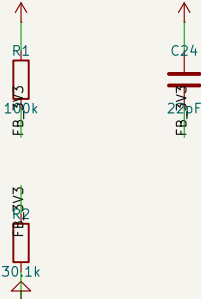
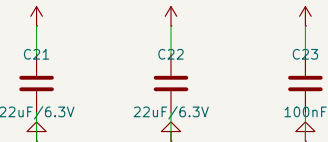
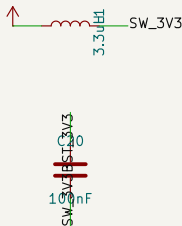
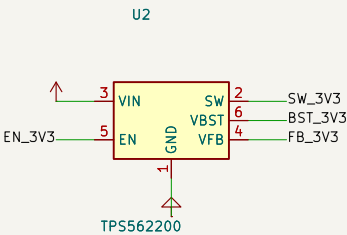
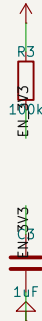
SDRAM balls come from fpga/spartan6/sp6_SDRAM.ucf; the rest live in scripts/gen_schematic.py. One run emits this drawing and the matching fpga/spartan6/singleboard.ucf.

SingleBoard project -- concept schematic, not fabricated		
Sheet: /		
File: singleboard.kicad_sch		
Title: SingleBoard 68K -- Overview		
Size: A3	Date: 2026-09-19	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/6

5 V input protection: polyfuse, series Schottky for reverse polarity,
TVS clamp. The Schottky costs ~0.4 V; both bucks accept 4.5 V minimum.

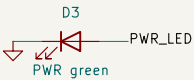
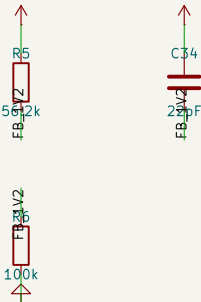
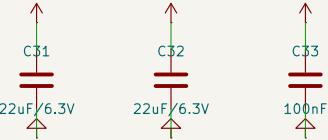
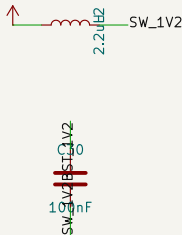
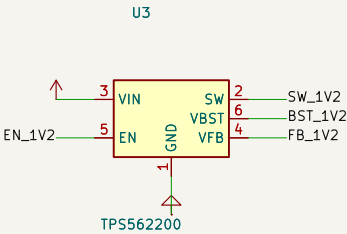


+3V3 buck -- $0.768\text{ V ref} \times (1 + 100\text{k}/30.1\text{k}) = 3.32\text{ V}$



R3/C3 delay the 3V3 enable ~100 ms behind 1V2.

+1V2 buck -- $0.768\text{ V ref} \times (1 + 56.2\text{k}/100\text{k}) = 1.20\text{ V}$



Estimated load: VCCINT ~0.3 A, VCCAUX+VCCO ~0.4 A, SDRAM ~0.15 A.

VCCAUX runs at 3.3 V, so the bitstream must carry CONFIG_VCCAUX = 3.3.

SingleBoard project -- concept schematic, not fabricated

Sheet: /power/
File: power.kicad_sch

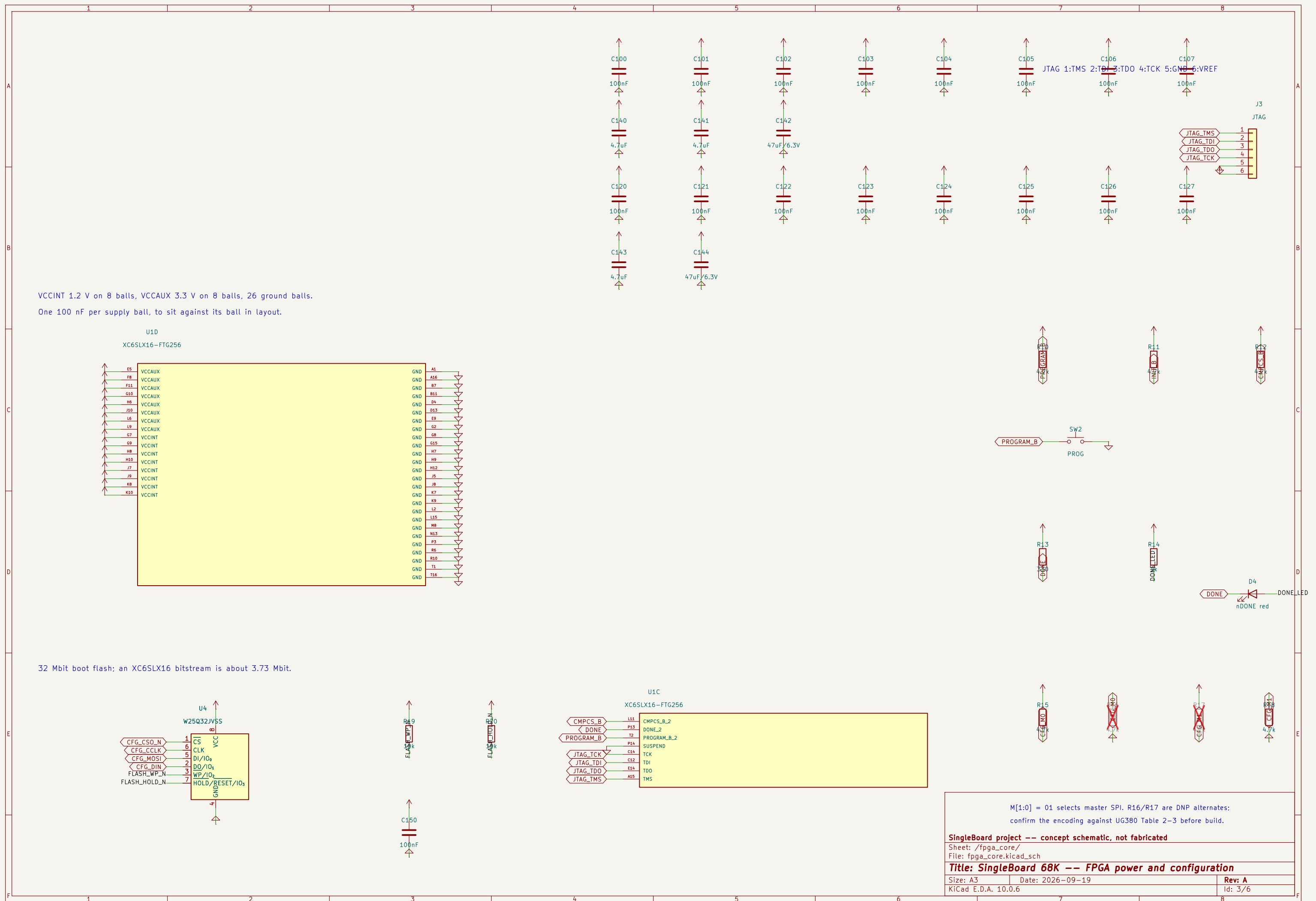
Title: SingleBoard 68K -- Power

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-19

Rev: A

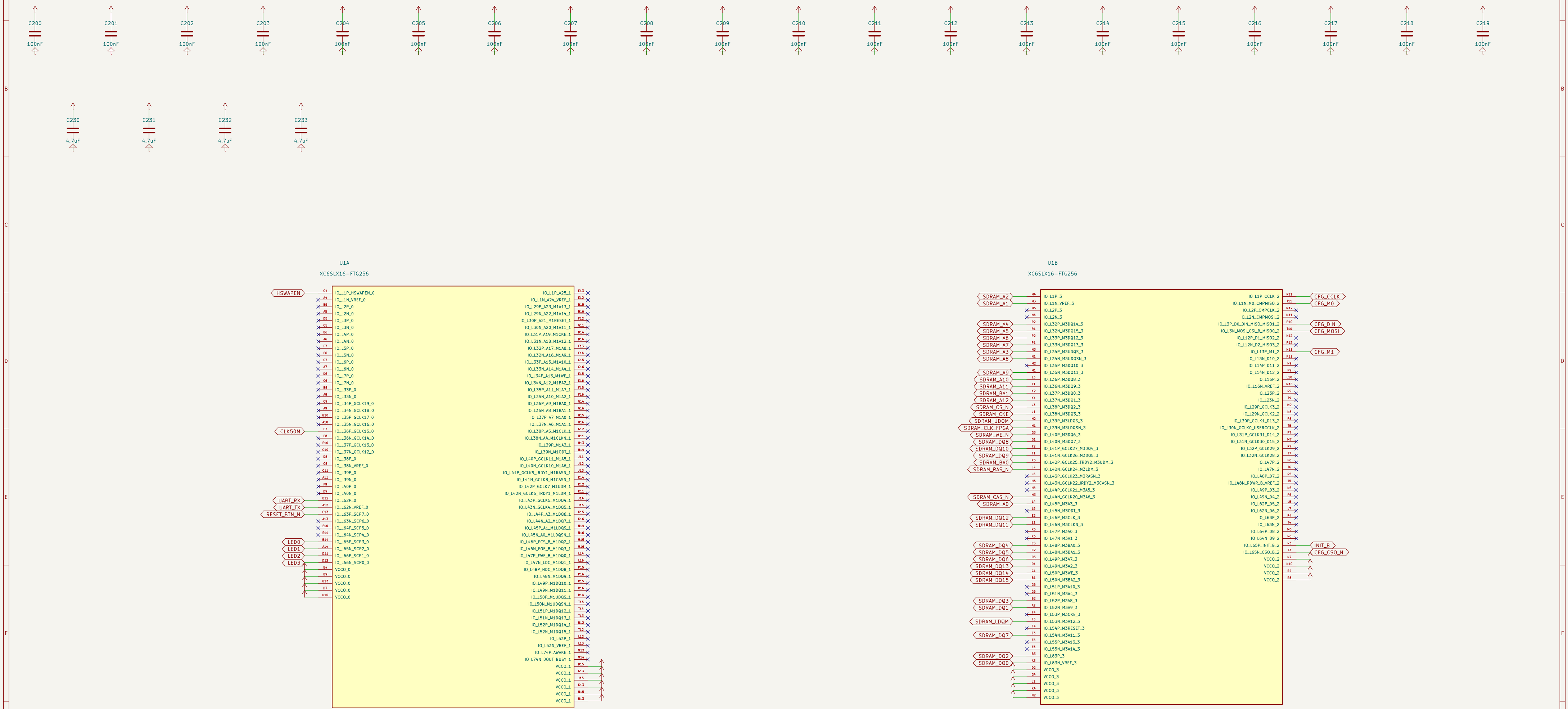
Id: 2/6



All four bank VCCO rails are 3.3 V. Unused bank pins carry a no-connect flag.

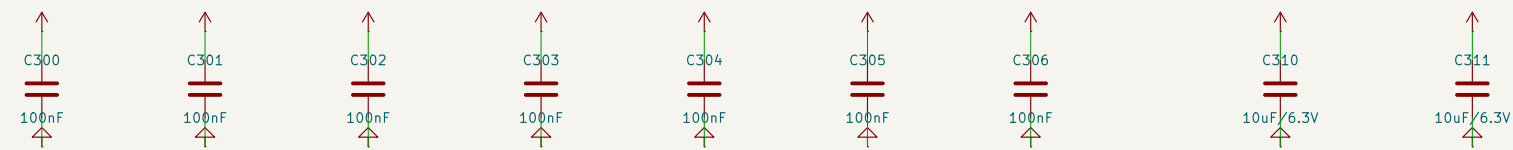
To bring a spare ball out later, delete its flag here and add a LOC to the UCF.

The SDRAM occupies bank 3, on the balls fpga/spartan6/sp6_SDRAM.ucf fixes; bank 1 is entirely spare.



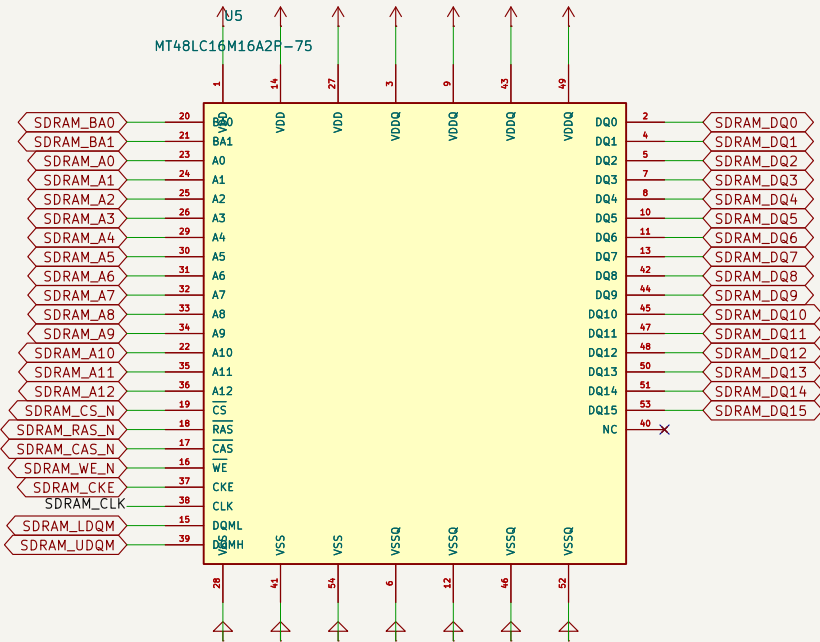
32 MiB: 4 banks x 8192 rows x 512 columns x 16 bits.

One 100 nF per VDD/VDDQ ball; VDD and VDDQ share the 3.3 V rail.



Ball assignment taken verbatim from fpga/spartan6/sp6_SDRAM.ucf.

R30 damps the forwarded clock. rtl/sdram_spartan6_clock.v drives SDRAM_CLK_FPGA from an ODDR2 with the clock inverted, so the device samples mid-cycle; see docs/sdram.md.



Pin numbers follow the JEDEC 54-pin TSOP-II arrangement -- verify against the Micron datasheet before fabrication.

SingleBoard project -- concept schematic, not fabricated

Sheet: /sdram/
File: sdram.kicad_sch

Title: SingleBoard 68K -- SDRAM

Size: A3

Date: 2026-09-19

Rev: A

KiCad E.D.A. 10.0.6

Id: 5/6

